GRouNdScale Size Comparison Report

Auto-generated from `size-compare` run artefacts

October 13, 2026

1 Overview

This report summarises 18 completed training runs of the causal GAN pipeline with increasingly higher highly variable gene (HVG) counts, from 991 to 17,789 genes. Every run uses the same architecture and training configuration (20 steps for the causal controller, then 20 steps for the main causal GAN); only the HVG count selected during preprocessing differs.

Causal-graph statistics are recomputed from the saved `causal_graph.pkl` of each run, because preprocessing and GRN inference are skipped when the graph already exists and the logs therefore contain no graph summary. The run requesting 18,000 HVGs resolved to 17,789 genes and is labelled by that gene count.

All runs completed without CUDA out-of-memory errors. The largest run peaked at 19,489 MiB, against a GPU capacity of 81,920 MiB.

2 GRN Characteristics

Table 1: Causal graph statistics per run. The TF/target ratio is transcription factors divided by target genes.

Genes	TFs	Target genes	TF/target (%)	Possible edges	Imposed edges	Density (%)
991	63	928	6.79	58,464	13,920	23.8
2,000	127	1,873	6.78	237,871	28,095	11.8
3,000	202	2,798	7.22	565,196	41,970	7.4
4,000	287	3,713	7.73	1,065,631	55,695	5.2
5,000	370	4,630	7.99	1,713,100	69,450	4.1
6,000	464	5,536	8.38	2,568,704	83,040	3.2
7,000	555	6,445	8.61	3,576,975	96,675	2.7
8,000	639	7,361	8.68	4,703,679	110,415	2.3
9,000	733	8,267	8.87	6,059,711	124,005	2.0
10,000	817	9,183	8.90	7,502,511	137,745	1.8
11,000	887	10,113	8.77	8,970,231	151,695	1.7
12,000	949	11,051	8.59	10,487,399	165,765	1.6
13,000	1,002	11,998	8.35	12,021,996	179,970	1.5
14,000	1,045	12,955	8.07	13,537,975	194,325	1.4
15,000	1,098	13,902	7.90	15,264,396	208,530	1.4
16,000	1,131	14,869	7.61	16,816,839	223,035	1.3
17,000	1,165	15,835	7.36	18,447,775	237,525	1.3
17,789	1,192	16,597	7.18	19,783,624	248,955	1.3

3 Training Step Times

Table 2: Average time per training step for the causal controller and the causal generator, taken from the final step log line of each phase.

Genes	Controller step (ms)	Generator step (ms)	TF/target (%)
991	33	81	6.79
2,000	37	149	6.78
3,000	35	221	7.22
4,000	40	284	7.73
5,000	19	348	7.99
6,000	38	352	8.38
7,000	36	455	8.61
8,000	37	512	8.68
9,000	39	594	8.87
10,000	37	643	8.90
11,000	38	698	8.77
12,000	33	772	8.59
13,000	37	816	8.35
14,000	37	888	8.07
15,000	42	952	7.90
16,000	36	1,019	7.61
17,000	39	1,095	7.36
17,789	42	1,201	7.18

Table 3: Expected training duration for 200,000 causal-controller steps and 500,000 causal-generator steps, extrapolated from the measured step times.

Genes	Controller (200,000 steps)	Generator (500,000 steps)	Total
991	1.8 h	11.2 h	13.1 h (0.5 d)
2,000	2.1 h	20.7 h	22.8 h (0.9 d)
3,000	1.9 h	30.7 h	32.6 h (1.4 d)
4,000	2.2 h	39.4 h	41.7 h (1.7 d)
5,000	1.1 h	48.3 h	49.4 h (2.1 d)
6,000	2.1 h	48.9 h	51.0 h (2.1 d)
7,000	2.0 h	63.2 h	65.2 h (2.7 d)
8,000	2.1 h	71.1 h	73.2 h (3.0 d)
9,000	2.2 h	82.5 h	84.7 h (3.5 d)
10,000	2.1 h	89.3 h	91.4 h (3.8 d)
11,000	2.1 h	96.9 h	99.1 h (4.1 d)
12,000	1.8 h	107.2 h	109.1 h (4.5 d)
13,000	2.1 h	113.3 h	115.4 h (4.8 d)
14,000	2.1 h	123.3 h	125.4 h (5.2 d)
15,000	2.3 h	132.2 h	134.6 h (5.6 d)
16,000	2.0 h	141.5 h	143.5 h (6.0 d)
17,000	2.2 h	152.1 h	154.2 h (6.4 d)
17,789	2.3 h	166.8 h	169.1 h (7.0 d)

4 Parameter Counts

Table 4: Parameter counts per model component and optimiser, in millions.

Genes	Generator	Critic	Labeler	Antilabeler	Gen. opt	Crit. opt	Labeler opt	Antilabeler opt
991	4.12M	1.67M	10.01M	10.01M	2.59M	5.02M	30.00M	30.00M
2,000	7.60M	2.71M	12.03M	12.03M	5.22M	8.12M	36.05M	36.05M
3,000	11.03M	3.73M	14.03M	14.03M	7.80M	11.19M	42.05M	42.05M
4,000	14.42M	4.75M	16.03M	16.03M	10.35M	14.26M	48.05M	48.05M
5,000	17.83M	5.78M	18.03M	18.03M	12.90M	17.33M	54.06M	54.06M
6,000	21.20M	6.80M	20.03M	20.03M	15.43M	20.40M	60.06M	60.06M
7,000	24.58M	7.83M	22.03M	22.03M	17.96M	23.48M	66.06M	66.06M
8,000	27.98M	8.85M	24.03M	24.03M	20.52M	26.55M	72.06M	72.06M
9,000	31.36M	9.87M	26.03M	26.03M	23.04M	29.62M	78.06M	78.06M
10,000	34.76M	10.90M	28.03M	28.03M	25.59M	32.69M	84.06M	84.06M
11,000	38.20M	11.92M	30.03M	30.03M	28.18M	35.76M	90.06M	90.06M
12,000	41.65M	12.95M	32.03M	32.03M	30.80M	38.84M	96.06M	96.06M
13,000	45.13M	13.97M	34.03M	34.03M	33.44M	41.91M	102.06M	102.06M
14,000	48.64M	14.99M	36.03M	36.03M	36.11M	44.98M	108.06M	108.06M
15,000	52.12M	16.02M	38.03M	38.03M	38.74M	48.05M	114.06M	114.06M
16,000	55.65M	17.04M	40.03M	40.03M	41.44M	51.12M	120.06M	120.06M
17,000	59.18M	18.07M	42.03M	42.03M	44.13M	54.20M	126.06M	126.06M
17,789	61.97M	18.87M	43.61M	43.61M	46.26M	56.62M	130.79M	130.79M

5 GPU Memory Requirements

Table 5: Peak GPU memory per training phase. Phase windows are taken from the training log timestamps and intersected with the `gpu-usage.csv` monitoring data.

Genes	Controller (MiB)	Generator (MiB)
991	6,079	7,551
2,000	6,119	8,223
3,000	6,161	8,993
4,000	6,229	9,757
5,000	6,293	10,239
6,000	6,373	10,965
7,000	6,413	11,691
8,000	6,453	12,461
9,000	6,513	13,173
10,000	6,355	13,843
11,000	6,645	14,617
12,000	6,699	15,249
13,000	6,751	16,037
14,000	6,843	16,755
15,000	6,895	17,511
16,000	6,947	18,215
17,000	6,999	18,949
17,789	7,015	19,489

6 Scaling Relations

Ordinary least-squares fits over all runs, with x the number of highly variable genes, give the causal-generator requirements

$$\widehat{M}_{\text{gen}}(x) = 0.7126\,x + 6778.8 \text{ MiB}, \qquad R^2 = 0.9997 \tag{1}$$

$$\widehat{t}_{\text{gen}}(x) = 0.06341\,x + 13.93 \text{ ms/step}, \qquad R^2 = 0.9961 \tag{2}$$

The memory fit uses the peak memory of the generator training window and the time fit uses the second final-step log line. The relations describe the measured range only.

6.1 Gene Count versus Target-Gene Count

Fitting against the number of target genes instead of highly variable genes gives $R^2 = 0.9998$ for memory and $R^2 = 0.9968$ for step time, compared with $R^2 = 0.9997$ and $R^2 = 0.9961$ for the gene-count fits. The two predictors are strongly correlated, so the differences are small, but the target-gene form is the better description of the underlying model size:

$$\widehat{M}_{\text{gen}}(y) = 0.7670\,y + 6805.3 \text{ MiB}, \qquad R^2 = 0.9998 \tag{3}$$

$$\widehat{t}_{\text{gen}}(y) = 0.06827\,y + 16.12 \text{ ms/step}, \qquad R^2 = 0.9968 \tag{4}$$

where y denotes the number of target genes in the causal graph.

7 Scaling Graphs

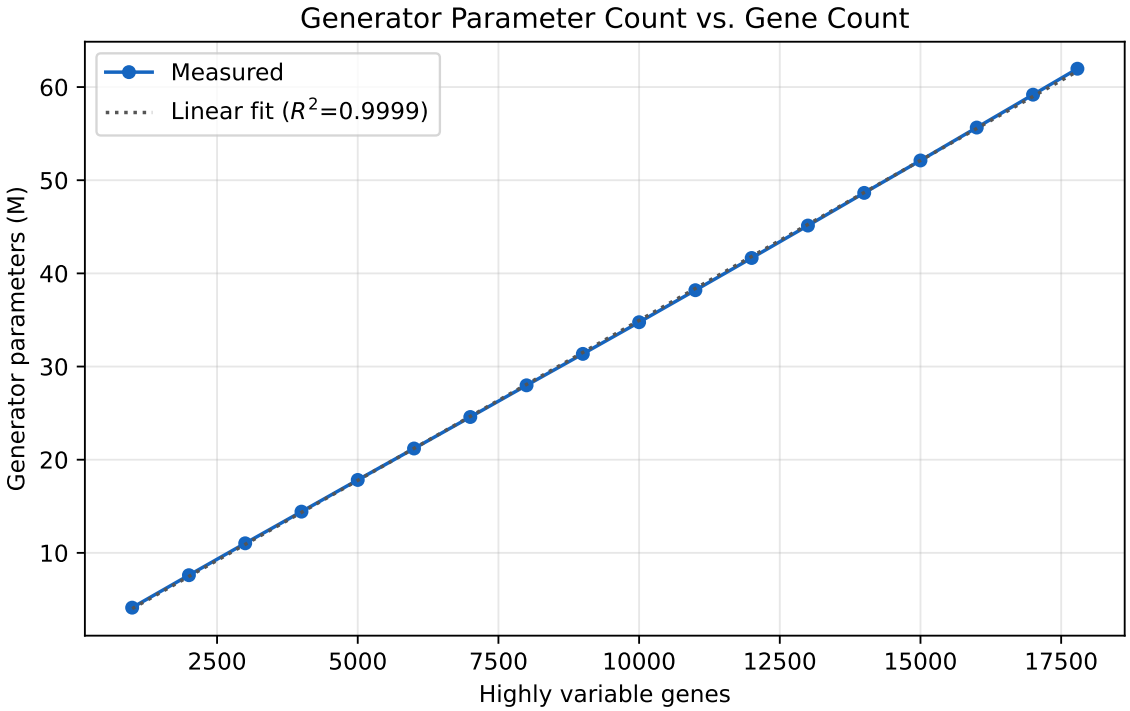


Figure 1: Generator parameter count versus gene count.

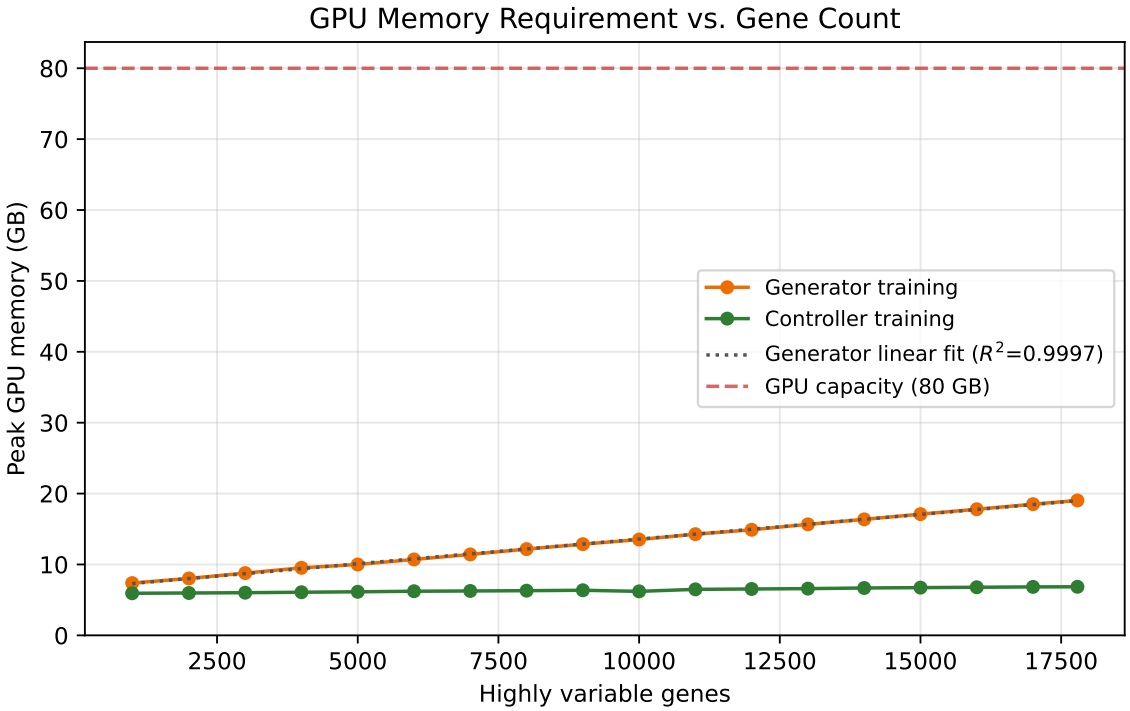


Figure 2: Peak GPU memory during controller and generator training versus gene count, with the linear fit and the GPU capacity.

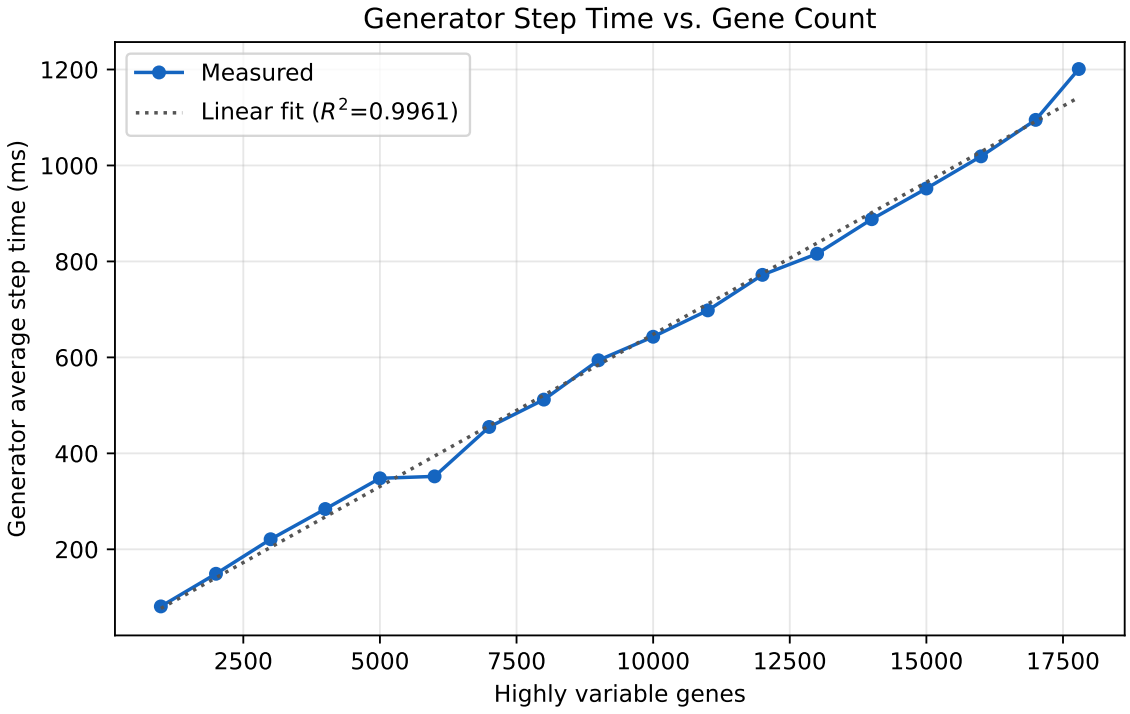


Figure 3: Causal-generator average step time versus gene count.

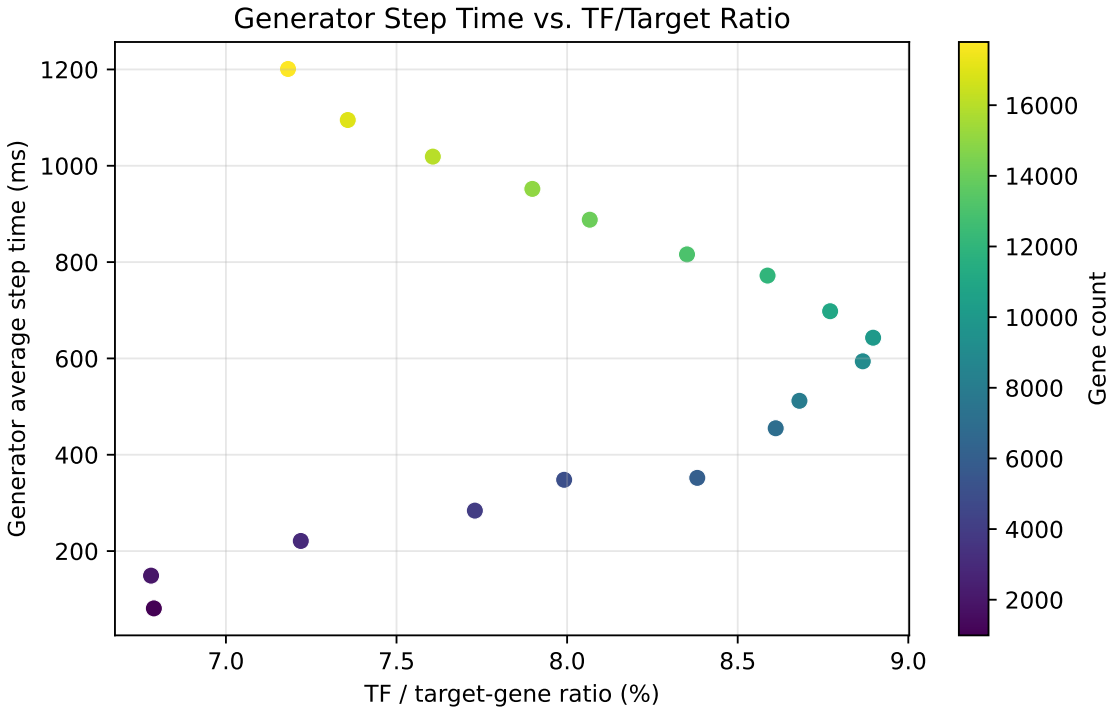


Figure 4: Causal-generator average step time versus the TF/target ratio; point colour encodes the gene count.

8 Notes

- All runs share the architecture: latent dim 128, generator layers 256–512–1024, critic layers 1024–512–256, labeler layers 2000–2000–2000, batch size 1024 and 20 steps per phase.
- The controller phase is trained first with batch size 128 and the main causal GAN second with batch size 1024; generator training dominates the memory requirement.
- Memory values are in MiB as reported by `nvidia-smi`.